

A TWO-PART INTERPRETABILITY STUDY

Interpreting Deepfake Detection on SID-Set

Foundation-Model Baselines and Mechanistic Analysis
via Sparse Autoencoders

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The challenge

Detecting real, synthetic, and tampered images on social media, three classes with very different structure

Two complementary lenses

Interpretable baselines on *frozen* foundation models + mechanistic analysis of the *fine-tuned* SOTA

Roadmap

- ① Problem, dataset, two-track approach
- ② **Part I:** Frozen-backbone baselines (CLIP, DINOv2, SigLIP + linear probes)
- ③ **Part II:** SIDA-7B through sparse autoencoders (Vanilla, TopK, Gated, JumpReLU, Matryoshka)
- ④ Synthesis: where the two lenses agree

The Deepfake Detection Challenge

Modern generators (FLUX, latent diffusion) produce images **indistinguishable from real** even to trained experts.

SID-Set frames detection as a **three-class** problem:

- **Real** — authentic photographs
- **Fully synthetic** — entirely AI-generated
- **Tampered** — real image, AI-inpainted region

Why three classes?

Binary (real vs. fake) collapses two fundamentally different regimes. Synthetic carries a *pervasive* generator signature; tampered is *mostly real* with only a local disruption.

Traditional detectors fail on SID-Set

Method	Acc.
FreDect	37.4%
CnnSpot	44.8%
Gram-Net	50.1%
LGrad	52.6%
UnivFD	55.3%
SIDA-7B (fine-tuned)	93.6%

Old artefact detectors don't generalise to modern diffusion models.

Dataset and Two-Track Approach

SID-Set (Huang et al., CVPR 2025)

Split	Real	Synth	Tamp
Part I train	2 667	2 667	2 666
Part I test	333	334	333
Part II train	10 000	10 000	10 000
Part II eval	10 000	10 000	10 000

Tampered images carry pixel-level masks. Real images from COCO / OpenImages / Flickr30k. Synthetic images from FLUX and latent diffusion.

Two complementary questions

Part I

What is already in
frozen VFM features?

shared benchmark

Part II

How does *fine-tuned*
SIDA-7B actually decide?

Both use SID-Set. Together they reach from raw pretrained features to inside the state-of-the-art.

PART I

Frozen Foundation-Model Baselines

What can simple linear probes on frozen vision features tell us about deepfakes?

Approach: Three Frozen Backbones + Linear Probes

Backbone	Training	Grid	Dim
CLIP ViT-B/32	Contrastive (V-L)	7×7	512
DINOv2-Base	Self-supervised	16×16	768
SigLIP-Base	Sigmoid contrastive	14×14	768

Pipeline

- Freeze backbone, extract global embedding
- Train L2/L1 logistic regression + decision tree
- Patch-token localizer for tampered regions
- 8 interpretability analyses on top

Design rationale

Freezing weights **isolates** what the pre-trained features already encode. All interpretability runs on cached features, **no GPU** at evaluation time.

Three paradigms covered: language-supervised (CLIP, SigLIP) and self-supervised (DINOv2), with no fine-tuning anywhere.

Classification Results

Backbone	Classifier	F1
CLIP ViT-B/32	LogReg-L2	0.890
	LogReg-L1 (277/512)	0.900
	DecTree (d=10)	0.691
DINOv2-Base	LogReg-L2	0.861
	LogReg-L1 (505/768)	0.867
	DecTree (d=10)	0.586
SigLIP-Base	LogReg-L2	0.857
	LogReg-L1 (324/768)	0.856
	DecTree (d=10)	0.663

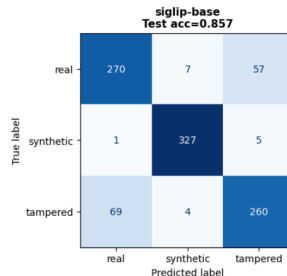
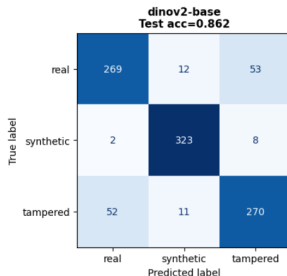
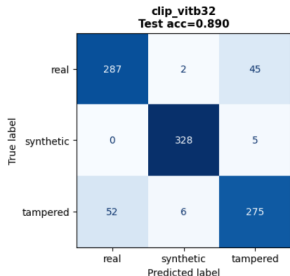
Three findings

- All backbones reach **86–90% F1** with a linear probe, no fine-tuning
- **L1 matches L2** while zeroing 34–58% of dims — heavy redundancy
- Trees collapse (59–69%): boundary is **linear but not axis-aligned**

CLIP at 90.0% F1 reaches 96% of fine-tuned SIDA-7B (93.5%), at a fraction of the cost.

Confusion Matrices: Tampered is the Hard Class

Confusion Matrices — L2 Logistic Regression (Test Set)



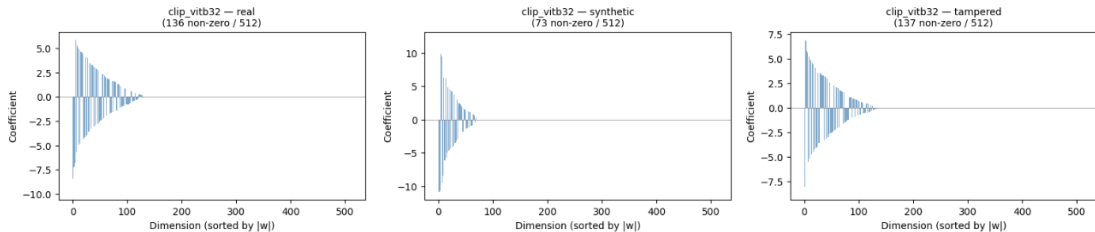
Across **all three backbones**, errors concentrate on the **tampered–real boundary**. Synthetic is almost perfect; real is strong; tampered lags.

Preview

This hard-tampered gap returns *sharply* in Part II.

L1 Sparsity: Where Each Class “Lives” in the Embedding

L1 Logistic Regression Coefficients (sorted by magnitude)



Asymmetric signatures

Synthetic: sparse, high-amplitude coefficients (up to +41.13 in DINOv2)

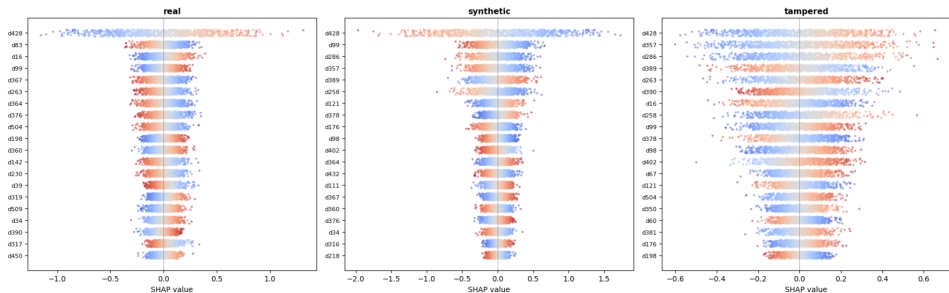
Tampered: distributed, moderate weights

Real: largely *negative* — defined by absence of forgery cues

Synthetic images produce a concentrated, easy-to-detect signal; tampered images require reading many dimensions at once.

SHAP Shortcut Analysis: Are These Spurious Features?

SHAP Summary - clip_vitb32 (Top-20 dims, test set)



Exact SHAP for a linear classifier: $\phi_j = w_j(x_j - \mathbb{E}[x_j])$. How many of the top-20 dims are *shared* across all three classes?

Backbone	Shared	R-only	S-only	T-only
CLIP	2	8	4	4
DINOv2	3	8	6	11
SigLIP	3	9	6	14

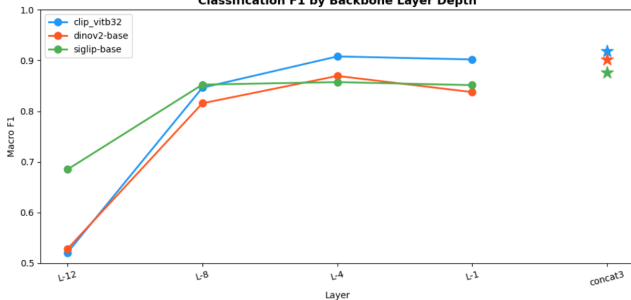
Not a shortcut

Only 2–3 of 20 top dimensions are shared across all classes. The classifier uses genuine *class-specific* cues, not a single spurious shortcut.

Dimension d428 (CLIP) appears across all classes but with *opposite signs* — the same feature, flipped per class. 10 / 29

Layer Probing: Where Do Deepfake Cues Emerge?

Classification F1 by Backbone Layer Depth



F1 vs. layer depth. Stars denote concat(L-1,-2,-3).

Backbone	L-12	L-4	concat
CLIP	0.521	0.908	<u>0.919</u>
DINOv2	0.528	0.870	<u>0.902</u>
SigLIP	0.685	0.857	<u>0.876</u>

Last quarter wins

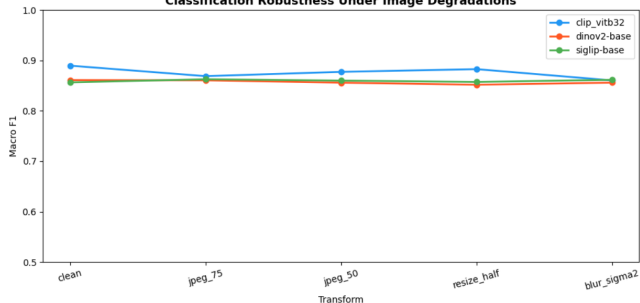
L-12 $\approx$ chance, **L-4** peaks.

For CLIP & DINOv2, L-4 *beats* the final layer L-1: late layers over-specialise for the pre-training objective.

Concat of last 3 layers: **+2–4 F1** free.

Robustness to Social-Media Transforms

Classification Robustness Under Image Degradations



Backbone	Clean	JPEG-50	Blur
CLIP	0.890	0.877	0.861
DINOv2	0.861	0.856	0.856
SigLIP	0.857	0.860	0.862

Remarkably stable

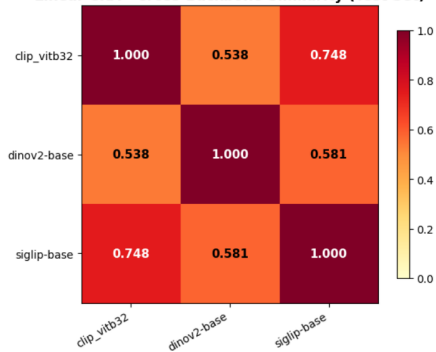
Worst degradation: -2.9 F1 (CLIP under blur).

DINOv2 is near-invariant (< 0.9 points).

Signal lives in high-level semantic features, not fragile frequency artefacts.

CKA: How Different Are the Three Representations?

Linear CKA - Cross-Backbone Similarity (Test Set)



	CLIP	DINOv2	SigLIP
CLIP	1.000	0.538	0.748
DINOv2	0.538	1.000	0.582
SigLIP	0.748	0.582	1.000

DINOv2 is the outlier

CLIP & SigLIP are similar (**0.748**): both use vision-language objectives.

DINOv2 (self-supervised) is distinct ($\leq \mathbf{0.58}$).

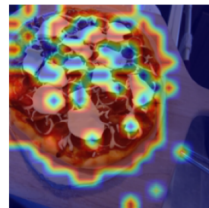
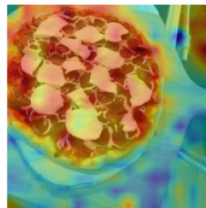
A CLIP+DINOv2 ensemble should capture complementary representations.

Localization: Spatial vs. Holistic Detection

Token-CAM vs Localizer — dinov2-base

GT mask token-CAM (IoU=0.61)

Localizer (IoU=0.67)



Patch localizer (dedicated):

	IoU	Dice	AUC
CLIP	0.375	0.504	0.805
DINOv2	0.323	0.441	0.803
SigLIP	0.279	0.403	0.771

Token-CAM (classifier attn.):

	IoU	Point.	AUC
CLIP	0.241	0.228	0.486
DINOv2	0.195	0.279	0.542
SigLIP	0.201	0.213	0.447

Holistic — not spatial

90% F1 classification **vs.** **only** 0.38 IoU localization.

Pointing accuracy **22–28%** $\approx$ chance.

The classifier detects tampering through a global embedding shift, not by localising the manipulated region.

Part I: Summary

What we learned about frozen features

- 86–90% F1 with simple linear probes, no GPU at eval
- Deepfake cues live in the **last quarter** of the backbone, not the final layer
- Class-specific features (SHAP) — **not** a shortcut
- Robust to JPEG, resize, blur (<3% drop)
- Decision is **holistic, not spatial** — classifier doesn't look *at* the tamper

Two findings that motivate Part II

1. Tampered is markedly harder than real or synthetic — why?
2. The decision is globally distributed across many dimensions — can we *decompose* it into interpretable features?

We now turn to the fine-tuned SIDA-7B model and decompose its internal representation with sparse autoencoders.

PART II

Mechanistic Analysis of SIDA via SAEs

Decomposing the 7B-parameter detector into interpretable, causally steerable features.

SIDA Architecture and Our Extraction Point

SIDA-7B pipeline

- 1 Image $\rightarrow$ CLIP encoder $\rightarrow$ image tokens
- 2 Image tokens + prompt $\rightarrow$ LLaMA-7B backbone
- CLS last-layer hidden state: $h_{\text{det}} \in \mathbb{R}^{4096}$
- 3 $h_{\text{det}} \rightarrow$ cls-head (Linear-ReLU-Linear) $\rightarrow$ 3 logits

What we extract

- 60 000 h_{det} embeddings (20k per class)
- Classification head weights (for interventions without re-running the model)

Class	SIDA Acc.
Real	98.4%
Full synthetic	99.9%
Tampered	82.5%
Overall	93.6%

Same asymmetry as Part I

Tampered is **16pp** behind real — scales up unchanged from the frozen-feature regime.

Five SAE Variants at a Glance

Variant	Sparsity mechanism
Vanilla	ReLU + L1 penalty ($\lambda=5\times 10^{-4}$)
TopK	Keep exactly top $k=64$ activations per sample
Gated	Sigmoid gate decides <i>whether</i> ; magnitude net decides <i>how much</i>
JumpReLU	Per-feature learnable threshold θ_i , straight-through estimator
Matryoshka	Nested widths $\{2048, 4096, 8192, 16384\}$; importance-ordered features

All SAEs: $d_{\text{in}}=4096 \rightarrow d_{\text{hidden}}=16,384$ ($4\times$ expansion), 50 epochs, Adam (lr 3×10^{-4}), decoder columns unit-normalised.

Why multiple variants?

Different sparsity mechanisms trade reconstruction fidelity against interpretability differently. We compare all five on **reconstruction**, **selectivity**, and **causal intervention** behaviour.

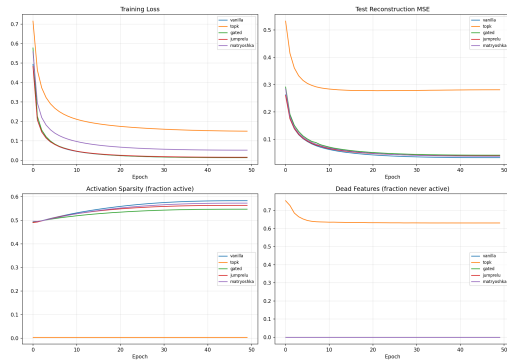
Reconstruction Quality and Training Dynamics

SAE	MSE	Exp. Var.	Dead
Vanilla	0.0328	96.7%	0%
TopK	0.2812	71.9%	63.1%
Gated	0.0419	95.8%	0%
JumpReLU	0.0392	96.1%	0%
Matryoshka	0.0379	96.2%	0%

TopK fails; others tie

TopK's 64-feature budget is **too tight** for faithful reconstruction and kills 63% of its features.

Four other variants all hit $\sim 96\%$ explained variance.



Training dynamics (50 epochs). TopK (orange) plateaus; others converge by epoch 30.

The SAE Preserves the Classification Signal

Round-trip test: $h_{\text{det}} \rightarrow \text{SAE} \rightarrow \text{decode} \rightarrow$
classification head

Method	All	Real	Synth	Tamp
Original (no SAE)	93.60	98.39	99.92	82.50
Vanilla	93.60	98.39	99.92	82.50
Gated	93.62	98.38	99.93	82.55
Matryoshka	93.58	98.32	99.92	82.51
JumpReLU	93.57	98.42	99.92	82.38
TopK	92.73	98.10	99.82	80.26

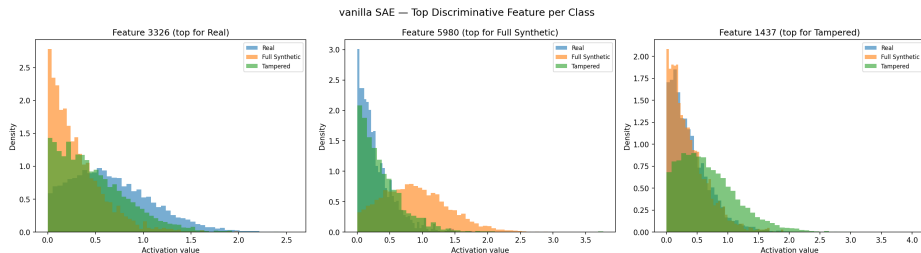
Nearly lossless

Vanilla / Gated / JumpReLU / Matryoshka:
<0.05 pp change.

Everything below is now trustworthy interpretability on the *actual* model.

TopK loses 2.24 pp on tampered — 64 features aren't enough to fully represent the tampered signal.

Feature Selectivity: Synthetic Features Are Specialists



Mean selectivity of top-20 features per class

SAE	Real	Synth	Tamp
Vanilla	0.476	0.787	0.515
TopK	0.959	1.000	0.959
Gated	0.483	0.867	0.524
JumpReLU	0.547	0.893	0.595
Matryoshka	0.446	0.916	0.494

Asymmetry echoes Part I

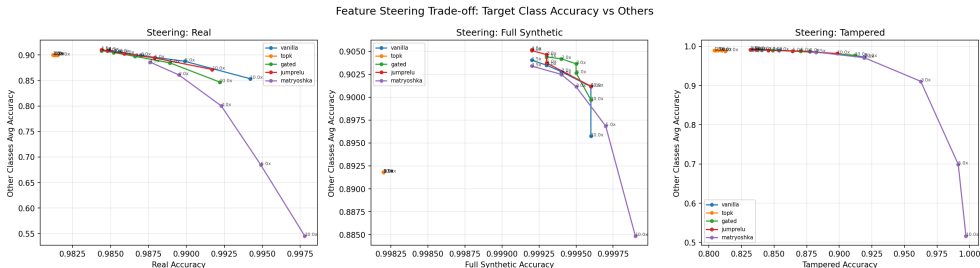
Synthetic: highly selective specialists (0.79–1.0)

Real & Tampered: more overlap, less class-specific

Exactly the pattern the L1 coefficients showed in Part I.

Feature 5980 (Synth) is highly bimodal and selective; Feature 1437 (Tamp) overlaps more with real, predicting the harder tampered class.

Causal Feature Steering: +9.4 pp on Tampered



Amplify top-20 tampered features by 10×

SAE	Tamp	ΔT	$\Delta Real$
Vanilla	91.92	+9.42	-3.72
Gated	91.20	+8.65	-2.87
JumpReLU	89.84	+7.46	-1.96
Matryoshka	99.72	+17.2	-94.8
TopK	81.27	+1.01	-0.34

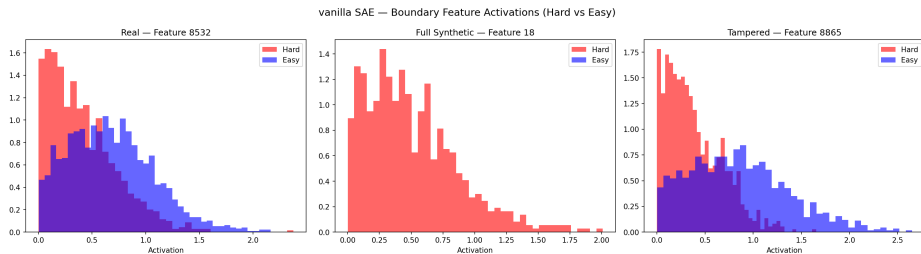
Causal control

Amplifying the right 20 features lifts tampered from **82.5** $\rightarrow$ **91.9%**.

A real knob, not a correlation.

Matryoshka over-steers: +17 pp tamp at the cost of -95 pp real — entangled features.

Decision Boundary: The Tampered Cliff



Class	Hard Acc.	Conf.	Easy Acc.
Real	91.95%	0.865	100%
Synthetic	99.33%	0.992	n/a
Tampered	12.70%	0.210	100%

Feature 8865 (tampered, $d=1.10$) cleanly separates *hard* (red) from *easy* (blue) samples — a candidate confidence calibration signal.

A severe failure mode

The bottom 20% of tampered images (by confidence) score just 12.7% — worse than random guessing among three classes.

≈4 000 tampered test images are effectively undetectable.

Mechanistic picture of SIDA-7B

- 4 of 5 SAEs reconstruct h_{det} near-losslessly (96.7% EV)
- Synthetic features are **highly selective** specialists
- Causal steering: **+9.4 pp tampered** accuracy (Vanilla, 10×)
- Tampered cliff**: hardest 20% collapse to 12.7%
- Synthetic detection: most accurate, *most fragile*

SAE variant verdict

Vanilla — best overall (fidelity + steerability)

JumpReLU — sparsest, minimal accuracy cost

Matryoshka — multi-scale but *entangled* (extreme steering)

TopK — too aggressive, loses information

Gated — competitive middle ground

Vulnerability map $\Rightarrow$ *concrete targets for adversarial hardening.*

SYNTHESIS

Where the Two Lenses Converge

Two methods, two model scales, one consistent picture of how deepfake detection works.

Four Bridges Between Part I and Part II

Bridge 1 Tampered is universally hard

Part I: worst class across 3 backbones.
Part II: 12.7% accuracy on hardest 20%.
Scale- and paradigm-independent.

Bridge 3 Holistic detection — not spatial

Part I: Token-CAM pointing $\approx$ chance.
Part II: per-class features distributed, not localised.
*The classifier doesn't look **at** the tamper.*

Bridge 2 Synthetic signal is concentrated

Part I: synthetic L1 coef up to +41 vs. ± 10 for real.
Part II: synthetic SAE selectivity 0.79–1.00.
Same asymmetry at every level.

Bridge 4 Class-specific — not shortcut

Part I: SHAP shared overlap 2–3 of 20.
Part II: steering one class disrupts the others.
Genuine per-class cues, not spurious shortcuts.

Where the Lenses Diverge (and Why It Matters)

Part I says

*Frozen features are **robust***

<3% F1 drop under JPEG-50 & blur

Signal = high-level semantic, not frequency artefacts

Part II says

*Fine-tuned features are **adversarially fragile***

Synthetic class: highest accuracy *and* highest vulnerability

Concentrated feature directions are attackable

Not a contradiction — two axes

Distributional robustness (benign corruption)

vs.

Adversarial robustness (targeted attack)

The features that make synthetic detection easy also make it *attackable*.

Deployment implication

Best picture may combine:

- the frozen regime's distributional robustness
- with vulnerability-aware adversarial training on the fine-tuned model's identified fragile directions.

What this study establishes

- ✓ Frozen VFM features already reach 96% of fine-tuned SOTA, interpretably and on CPU
- ✓ First SAE-based mechanistic analysis of a multimodal deepfake detector
- ✓ Causal feature steering: a real knob (+9.4 pp tampered)
- ✓ Two independent analyses that converge on the same qualitative picture

Open directions

- Extend SAE analysis to *intermediate* LLM layers (logit-lens across depth)
- Correlate SAE features with *spatial* tampered-region properties
- Vulnerability-guided adversarial training on the top-20 fragile directions per class
- Cross-backbone ensembles informed by CKA (DINOv2 + a VLM)

Thank you

Questions?

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